Fiscal Year:	FY 2014	**Task Last Updated:**	FY 10/29/2014
PI Name:	Thomopoulos, Stavros Ph.D.		
Project Title:	The effect of weightlessness on the tendon-to-bone insertion		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	thomopouloss@wudosis.wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314-362-8605
Organization Name:	Washington University		
PI Address 1:	Department of Orthopaedic Surgery		
PI Address 2:	425 S Euclid Ave, Box 8233		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63110-1010	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 , STS-133, STS-135, Bion M1 NOTE: Extended to 7/31/14 per A. Chu/ARC and NSSC info (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AP05G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Soft tissue injuries often occur at the sites of attachment of tendons and ligament to bone, the so-called "insertion sites". The sensitivity of the musculoskeletal system to its mechanical environment may magnify the injury susceptibility of insertion sites during prolonged exposure to microgravity. Changes in loading across joints, as would occur during a long stay on the International Space Station or during a trip to Mars, will lead to changes in tissue mechanical, structural, and biochemical properties. Joint unloading will result in a catabolic biologic state and subsequent tissue degeneration. Physiologic joint loading, on the other hand, will result in an anabolic biologic state and maintenance of tissue properties. Joint deterioration during space flight would significantly increase the risk of injury to the tendons and ligaments which provide function and stability to joints. These injuries do not heal well (e.g., the failure rate for repaired rotator cuff repairs may be as high as 94%) and will severely debilitate an injured astronaut. The negative effect of unloading has most clearly been demonstrated in bone and muscle. Less is known, however, about the effect of weightlessness on joint soft tissues. Therefore, the overall objective of this proposal is to examine the effect of prolonged weightlessness on the mechanical and biochemical properties of muscles, tendons, and their insertions into bone. We are studying the tendon-to-bone insertions of animals subjected to 14 – 30 days of weightlessness through the Biospecimen Sharing Program (BSP) organized for the STS-131, STS-133, STS-135, and the Bion-M1 missions. These shared animals allow us to examine joint degeneration due to weightlessness by comparing the weightless group to a control group on Earth. Outcome measures include assessment of the mechanical, structural, genetic, and compositional properties of mouse tendon-to-bone insertions. These analyses are performed on mice whose procedures and objectives were associated with the primary research focus of this research program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Musculoskeletal injuries are a common cause of pain and disability, and result in significant health care costs. Approximately 30% of work related injuries in the United States each year are to the upper extremity. At the rotator cuff, tendon-to-bone repair techniques have been plagued by a high rate of recurrent tears, ranging from 20-94%. There is little scientific basis for the current clinical management of this injury. Regeneration of the insertion site of the tendon is critical for the restoration of tendon function and for the prevention of re-injury. However, clinical and animal model studies have demonstrated a lack of healing potential after tendon-to-bone injury and repair. Musculoskeletal tissues are sensitive to their mechanical environment. Increased loading (e.g., through exercise) generally leads to an increase in mechanical properties, while decreased loading (e.g., through immobilization) generally leads to decreases in mechanical properties. For example, bone adapts to its loading environment by remodeling its architecture to optimally transfer stresses. Unloading leads to rapid resorption of bone and a decrease in bone mineral density. Increased loading, on the other hand, promotes new bone formation and an increase in bone mineral density. These changes are especially evident at the bony end of the tendon-to-bone insertion. Understanding the responses of tendons and their bony insertions to a weightless environment will allow us to develop strategies for preventing tendon-to-bone injuries and/or improving the healing response of tendon-to-bone injuries for humans on Earth.
Task Progress:	SUMMARY OF PAST YEAR'S RESEARCH ACCOMPLISHMENTS Muscles, tendons, and bony insertions of animals subjected to 13-30 days of weightlessness on shuttle missions STS-131 and STS-135 (13-15 days microgravity + ~3 hours recovery), and Bion-M1 (30 days microgravity + ~13 hrs recovery) were studied. We completed the following in the fifth year of the grant: assessment of rotator cuff muscles/tendons/insertions from STS-131/135 and Bion-M1 and further assessment of Achilles tendons/insertions from Bion-M1. Specimens from STS-133 are currently being stored for future analysis. Bion-M1 launched on April 19, 2013 and landed on May 19, 2013. A number of animals flew on this mission; 6 of the mice on the satellite were allocated to the NASA BSP investigators. An additional 8 mice were housed at the Russian Federation Institute for Biomedical Problems (IMBP) to serve as ground vivarium controls. Left and right shoulders (humerus-rotator cuff-scapula units) and ankles (Achilles tendon-calcaneous bone) were removed from each mouse approximately 12 hours after landing, processed, and shipped to our laboratory. Ongoing studies analyzing specimens from this and previous missions include: proteomics on rotator cuff mouse muscles, proteomics of hind paw flexor and extensor tendons, micro-computed tomography of the humeral head and calcaneal bones, biomechanics of the supraspinatus and Achilles tendon-to-bone insertions, and histology of supraspinatus tendon-humeral head and Achilles tendon-calcaneous samples. 1. The effect of microgravity on tendons and their bony insertions For tendons and their bony insertions, enzymatic degradation of the tissue by matrix metalloproteinases (MMPs) was evident after tendon unloading and/or injury. Microgravity led to significant increases in MMP-3 and MMP-13 mRNA expression (data presented in previous progress report) for STS-131 supraspinatus samples (Bion M1 samples are currently being analyzed). This degradation was consistent with decreased collagen fibril diameters in tendons from Bion samples and decreased supraspinatus tendon cross-sectional areas from both STS-131/135 and Bion M1 samples. Surprisingly, decreased cross-sectional area did not result in changes in the force-deformation behavior of the tissue (e.g., failure force). When normalizing mechanical properties for cross-sectional area and length (i.e., stress-strain properties), an increase in the modulus and failure stress (strength) of the tendon-to-bone insertion was seen for both STS and Bion samples. The stiffer tendon-to-bone insertion may be indicative of early tissue pathology, as is seen in glycation-related stiffening of soft tissues in a number of diseases. 2. The effect of microgravity on bone Joint unloading on Earth or in space has been shown to cause a rapid loss of bone mineral mass. Changes in mineral are typically due to increased bone resorption by osteoclasts, resulting in decreased mineral content as well as changes in the trabecular bone architecture. It is likely that similar bony changes will occur at tendon-to-bone insertions subjected to a weightless environment. Data from the STS-131 and STS-135, presented in the previous progress report, demonstrate that bone resorption is increased (as indicated by increases in the ratio of RANK-L to OPG gene expression) and osteoblast differentiation is decreased (as indicated by decreases in osterix and Runx-2 gene expression)). Bone mineral density (BMD) and trabecular thickness (Tb.Th.) were decreased due to microgravity. We are currently evaluating the gene expression and bony architecture or humeral heads and calcaneal bones from Bion animals. 3. The effect of microgravity on the rotator cuff muscles Contrasting results were seen when comparing gene and protein expression from STS and Bion flights. Microgravity significantly downregulated the expression of both myogenic and adipogenic genes after 15 days of microgravity + ~3 hrs of recovery in STS-131 animals (note: "recovery" indicates time elapsed from landing to sacrifice). The mRNA

	changes measured in myogenic factors indicated atrophy/degeneration of the muscles due to microgravity. Downregulation of genes related to the adipogenic program indicated that degeneration due to microgravity may occur through different pathways than degeneration due to tendon and/or nerve injury in gravity conditions. In contrast, opposite trends (or no effect) were seen after 30 days of microgravity + ~13 hrs of recovery in Bion-M1 animals. Proteomics analysis of muscles showed similar results. The gene and protein expression changes seen in the Bion-M1 muscles is likely due to initiation of a recovery phase after return of gravity loading. Expression changes can occur within hours of a loading event, whereas mass and functional changes may take days or weeks to manifest. We are completing the analysis of Bion-M1 samples and preparing two manuscripts on the results.
Bibliography Type:	Description: (Last Updated: 10/19/2020)
Abstracts for Journals and Proceedings	Thomopoulos S. "The Effects of Microgravity on the Rotator Cuff of the Shoulder." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Thomopoulos S, Fitzgerald J. "Workshop: New insights into the effects of spaceflight in musculoskeletal tissues." Orthopaedic Research Society Annual Meeting 2012, San Francisco, CA, February 4–7, 2012. Orthopaedic Research Society Annual Meeting 2012, San Francisco, CA, February 4–7, 2012. , Feb-2012
Articles in Peer-reviewed Journals	Shen H, Lim C, Schwartz AG, Andreev-Andrievskiy A, Deymier AC, Thomopoulos S. "Effects of spaceflight on the muscles of the murine shoulder." FASEB J. 2017 Dec;31(12):5466-77. Epub 2017 Aug 17. https://doi.org/10.1096/fj.201700320R ; PubMed PMID: 28821629; PubMed Central PMCID: PMC5690377 , Dec-2017
Articles in Peer-reviewed Journals	Deymier AC, Schwartz AG, Lim C, Wingender B, Kotiya A, Shen H, Silva MJ, Thomopoulos S. "Multiscale effects of spaceflight on murine tendon and bone." Bone. 2020 Feb;131:115152. https://doi.org/10.1016/j.bone.2019.115152 ; PMID: 31730829; PMCID: PMC7138367 , Feb-2020